

Surrogate Observability Pattern

direct_vs_middleware: client-side differential experiment

Authorised evidence boundary: RDG timestamps, outcomes, and correlation metadata only

PATH A — DIRECT

RDG
client timestamps

direct interaction

Downstream Service
target operation

$L_A(q)$: client-observed end-to-end latency

PATH B — MEDIATED

RDG
client timestamps

INTERNAL LOGS UNAVAILABLE

Gateway (APMS)
protected middleware layer

Downstream Service
same target operation

$L_B(q)$: client-observed end-to-end latency

Surrogate differential estimate

$$\Delta L(q) = L_B(q) - L_A(q) \quad q \in \{\text{mean}, P_{50}, P_{95}, P_{99}\}$$

No middleware timestamps required: ΔL estimates apparent mediated-path overhead

Comparable workload, downstream service, load profile, and observation boundary are required.

It isolates the aggregate mediation effect, **not** APMS internal processing time; network routing, security processing, transformations, and retries may also contribute to ΔL .